

Physics-Constrained Multimodal Retrieval of Lunar Elemental Ratios

A Research-Grade Methodology for Chandrayaan-2 IIRS and CLASS-XRF Data Fusion

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Abstract

Near-infrared (NIR) spectroscopy and X-ray fluorescence (XRF) provide complementary measurements of lunar composition. NIR observations from instruments such as the Chandrayaan-2 Imaging Infrared Spectrometer (IIRS) primarily constrain mineralogy, mineral chemistry, grain size, space-weathering state, and vibrational/electronic absorption features, whereas the Chandrayaan-2 Large Area Soft X-ray Spectrometer (CLASS) directly measures elemental X-ray fluorescence lines for major elements under suitable solar illumination. Consequently, elemental ratios such as Mg/Si, Al/Si, Ca/Si, and Fe/Si should not be interpreted as native NIR observables.

This methodology formulates the problem as a multimodal inverse problem. IIRS spectra are converted into calibrated reflectance and then analyzed using classical absorption parameters, nonlinear radiative-transfer spectral unmixing, and physics-aware machine-learning representations. CLASS spectra are independently processed through solar-state screening, particle-background subtraction, instrumental response correction, and XRF line fitting. The two data products are subsequently harmonized in spatial scale and fused using Bayesian or physics-constrained machine-learning inference. Grain size, space weathering, mineral mixing, photometric geometry, thermal emission, calibration uncertainty, and model discrepancy are explicitly treated as nuisance or latent variables.

The final data product consists of elemental ratios with uncertainty distributions, detection limits, quality flags, provenance, and validation pedigree. NIR-only products are restricted to mineral-chemical quantities such as Mg#, Fo#, En#, Wo#, modal mineral fractions, and FeO proxies. Absolute elemental ratios are reported only after calibration or fusion with CLASS-XRF and independent geochemical constraints.

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1 Scientific Scope and Measurement Philosophy

1.1 Problem Definition

The objective is to retrieve lunar elemental ratios from complementary hyperspectral and XRF measurements while preserving the distinction between directly measured quantities and model-dependent proxies.

For the NIR measurement, the forward model is written as

$$\mathbf{y}_{\text{NIR}} = \mathcal{F}_{\text{NIR}}(\mathbf{m}, \mathbf{g}, \mathbf{w}, \mathbf{p}, T) + \epsilon_{\text{NIR}}, \quad (1)$$

where $\mathbf{m}$ represents modal mineral abundances, $\mathbf{g}$ mineral chemistry, $\mathbf{w}$ space-weathering state, $\mathbf{p}$ photometric geometry, T surface temperature, and ϵ measurement noise.

For CLASS-XRF,

$$\mathbf{y}_{\text{XRF}} = \mathcal{F}_{\text{XRF}}(\mathbf{c}, \mathbf{G}, \mathbf{S}, \mathbf{B}) + \epsilon_{\text{XRF}}, \quad (2)$$

where $\mathbf{c}$ is elemental abundance, $\mathbf{G}$ observation geometry, $\mathbf{S}$ the incident solar spectrum/state, and $\mathbf{B}$ the instrumental and particle background.

The desired posterior is therefore

$$P(\mathbf{m}, \mathbf{g}, \mathbf{c}, \mathbf{w} \mid \mathbf{y}_{\text{NIR}}, \mathbf{y}_{\text{XRF}}). \quad (3)$$

This formulation is preferable to a simple regression from NIR band parameters to elemental ratios because it explicitly represents the degeneracies inherent in reflectance spectroscopy.

1.2 Measurement Complementarity

NIR absorptions principally arise from crystal-field transitions, charge-transfer processes, and molecular vibrational overtones/combination bands. They therefore constrain mineralogical state rather than directly counting atoms.

CLASS-XRF, by contrast, provides direct information on characteristic X-ray fluorescence lines. The methodological distinction is:

$$\boxed{\text{NIR} \rightarrow \text{mineralogy and mineral chemistry}} \quad (4)$$

$$\boxed{\text{CLASS-XRF} \rightarrow \text{elemental abundance}} \quad (5)$$

The research problem is consequently a *data-fusion inverse problem*, not merely an NIR elemental-ratio regression.

2 Data Sources and Instrument-Specific Products

2.1 Chandrayaan-2 IIRS

IIRS data should be treated as a calibrated hyperspectral data cube rather than a collection of isolated band measurements. The operational chain should preserve the complete spectrum wherever possible.

The useful spectral regimes are divided into:

- a. approximately 0.8–2.5 μm : primary mineralogical VNIR/SWIR regime;
- b. approximately 2.5–3.3 μm : strongly coupled mineralogical, hydration, and thermal regime;
- c. approximately 3.3–5.0 μm : increasingly important thermal/emissive regime.

Recent IIRS Level-2 processing work emphasizes radiometric correction, thermal correction, apparent reflectance, standardized geometry, empirical-line correction, and validation against independent lunar observations. These steps should be reflected explicitly in the pipeline.

2.2 CLASS-XRF

CLASS measurements should be processed as event/spectrum data with explicit solar-state and particle-background quality control. Elemental inference should be based on physically modeled characteristic lines rather than simple raw count ratios.

Relevant characteristic lines include Mg, Al, Si, Ca, and Fe fluorescence features. The final XRF elemental retrieval should incorporate:

- detector response;
- energy calibration;
- line-spread function;
- solar spectral state;
- continuum background;
- particle-induced background;
- pile-up and instrumental artifacts;
- fluorescence line fitting;
- uncertainty in fundamental-parameter conversion.

3 IIRS Level-1 to Level-2 Processing

3.1 Detector and Radiometric Calibration

Starting from detector digital numbers,

$$DN_1 = DN_{\text{raw}} - D(T_{\text{det}}) t_{\text{exp}} - B_{\text{bias}}, \quad (6)$$

followed by flat-field correction,

$$DN_2 = \frac{DN_1}{F_{\text{flat}}(x, \lambda)}. \quad (7)$$

Bad pixels, dropouts, cosmic-ray spikes, nonlinear detector response, and saturation should be flagged rather than silently interpolated.

Radiance is obtained through an instrument calibration function,

$$L_{\text{rad}}(\lambda) = DN_2 G(\lambda), \quad (8)$$

where $G(\lambda)$ incorporates the relevant calibration coefficients.

3.2 Wavelength Calibration

The wavelength solution may initially be represented as

$$\lambda_0(p) = a_0 + a_1p + a_2p^2 + \dots, \quad (9)$$

with detector temperature, optical flexure, smile, and spatially dependent wavelength offsets included in the operational calibration.

The corrected spectrum may be resampled onto a common wavelength grid using a flux-preserving interpolation scheme.

The residual wavelength error should be propagated:

$$\sigma_{\lambda,\text{total}}^2 = \sigma_{\lambda,\text{fit}}^2 + \sigma_{\lambda,\text{thermal}}^2 + \sigma_{\lambda,\text{cal}}^2. \quad (10)$$

3.3 Thermal Emission Removal

For wavelengths where thermal emission becomes significant,

$$L_{\text{obs}} = L_{\text{reflected}} + \epsilon(\lambda)B_{\lambda}(T), \quad (11)$$

where $B_{\lambda}(T)$ is the Planck function.

Thermal removal should be iterative because errors in T and emissivity propagate strongly into the 2.5–5 μm region.

The retrieved temperature should therefore be treated as an inferred nuisance parameter with uncertainty rather than as an exact external quantity.

3.4 Photometric Correction

The reflectance should be normalized to a common geometry using an appropriate photometric model such as Hapke, Lommel–Seeliger, Akimov, or an empirically validated lunar disk function.

$$R_{\text{std}}(\lambda) = \mathcal{P} [R_{\text{obs}}(\lambda); i, e, g \rightarrow i_0, e_0, g_0]. \quad (12)$$

The photometric correction must not be assumed to remove space-weathering effects.

4 Spectral Feature Extraction

4.1 Continuum Removal

For an absorption feature,

$$R_n(\lambda) = \frac{R(\lambda)}{R_c(\lambda)}, \quad (13)$$

where R_c is a locally estimated continuum.

Band depth is

$$BD(\lambda_c) = 1 - R_n(\lambda_c). \quad (14)$$

Equivalent width is

$$EW = \int_{\lambda_1}^{\lambda_2} [1 - R_n(\lambda)] d\lambda. \quad (15)$$

Band area may be used to calculate

$$BAR = \frac{BA_{II}}{BA_I}. \quad (16)$$

Continuum methods should include convex-hull, polynomial, and physically motivated local baselines. The difference between methods is itself a systematic uncertainty.

4.2 Classical Mineralogical Parameters

The principal parameters include

$$\lambda_{BI}, \quad \lambda_{BII}, \quad BD_I, \quad BD_{II}, \quad BA_I, \quad BA_{II}, \quad BAR,$$

along with albedo, continuum slope, spectral curvature, and hydration-region parameters.

These quantities should be retained even when a full-spectrum machine-learning representation is subsequently used.

4.3 Full-Spectrum Representation

The complete spectrum should also be represented using methods such as:

- PCA/MNF;
- wavelet representations;
- autoencoder embeddings;
- physically constrained latent representations;
- nonlinear radiative-transfer parameters.

This avoids the information loss associated with reducing every spectrum to a small number of band parameters.

5 Nonlinear Spectral Unmixing

5.1 Motivation

Linear spectral mixing is a useful baseline, but lunar regolith is an intimate particulate mixture for which nonlinear scattering can be important. The final inversion should therefore compare linear mixing with nonlinear radiative-transfer models.

A generic physical mixture model is

$$R_{\text{model}}(\lambda) = \mathcal{H}(f_1, \dots, f_N, D, \theta, w), \quad (17)$$

where f_j are modal abundances, D represents grain-size parameters, θ represents photometric/radiative-transfer parameters, and w represents optical maturity or space-weathering state.

5.2 Hapke/Shkuratov Inversion

The inversion minimizes

$$\chi^2 = \sum_{\lambda} \frac{[R_{\text{obs}}(\lambda) - R_{\text{model}}(\lambda)]^2}{\sigma_R^2(\lambda)}. \quad (18)$$

Constraints include

$$f_j \geq 0, \quad (19)$$

and

$$\sum_j f_j = 1. \quad (20)$$

The solution should be reported as a posterior distribution rather than as a single deterministic vector whenever computationally feasible.

5.3 Machine-Learning Surrogate

Recent research combining nonlinear Shkuratov modeling with machine learning provides a useful framework for scaling physically based inversion to large hyperspectral datasets.

A surrogate model may be defined as

$$\mathbf{z} = NN_\theta [R(\lambda)], \quad (21)$$

where $\mathbf{z}$ contains mineral fractions and mineral-chemical parameters.

The ML model should be trained on synthetic spectra generated from the physical forward model, supplemented by laboratory and flight observations.

6 Explicit Treatment of Space Weathering and Grain Size

Space weathering should be treated as a latent variable rather than simply removed as a preprocessing nuisance.

The reflectance model is therefore

$$R(\lambda) = F(\text{mineralogy, chemistry, grain size, maturity, geometry}). \quad (22)$$

A useful parameter vector is

$$\boldsymbol{\theta} = [f_{\text{ol}}, f_{\text{opx}}, f_{\text{cpx}}, f_{\text{plg}}, \text{Fo}\#, \text{En}\#, \text{Wo}\#, D, \tau_{\text{weathering}}]. \quad (23)$$

Inference should marginalize over D and $\tau_{\text{weathering}}$ rather than assuming they are known.

This is essential because grain size, optical maturity, albedo, continuum slope, and absorption strength can produce degeneracies that otherwise appear as compositional trends.

7 NIR-Derived Mineral Chemistry and Proxy Products

7.1 Olivine

Olivine composition should primarily be expressed using Fo# or Fa# rather than directly claiming a measured Mg/Si ratio.

$$\text{Fo}\# = 100 \frac{\text{Mg}}{\text{Mg} + \text{Fe}}. \quad (24)$$

The conversion from spectral band position to Fo# should be treated as a calibration relation with uncertainty, not a universal exact equation.

7.2 Pyroxene

For pyroxene, the inversion should estimate En, Fs, and Wo components where the calibration permits:

$$En + Fs + Wo = 100. \quad (25)$$

Band I and Band II centers, areas, widths, and asymmetries should be used jointly.

7.3 Feldspathic Material

Plagioclase is difficult to quantify from NIR alone because relatively Fe-poor feldspar can have weak absorptions.

Consequently, NIR should report:

- feldspathic probability;
- modal plagioclase estimate with uncertainty;
- anorthite-related classification where justified;
- confidence/upper-limit flags.

Al/Si should not be treated as a direct NIR measurement.

8 Elemental-Ratio Feasibility

Table 1: Recommended interpretation of NIR and CLASS observables.

Ratio	NIR information	NIR interpretation	CLASS interpretation
Mg/Si	Olivine/pyroxene chemistry, Mg#, Fo#, En#	Model-dependent proxy	Direct XRF elemental constraint
Fe/Si	Band strengths, Fe-bearing mineral abundance, FeO proxy	Strongly affected by maturity and grain size	Direct XRF elemental constraint
Ca/Si	Pyroxene Wo#, HCP/LCP discrimination	Mineralogical proxy; plagioclase correction required	Direct XRF elemental constraint
Al/Si	Feldspathic probability, weak Fe-bearing feldspar feature	Low-precision indirect proxy	Direct XRF elemental constraint

NIR-only elemental ratios should therefore be labeled explicitly as *model-dependent reconstructed ratios*. True elemental ratios should be produced only after external calibration or multimodal fusion.

9 CLASS-XRF Processing

9.1 Solar-State Screening

CLASS XRF intensity is strongly dependent on incident solar X-ray flux. Observations should therefore be classified according to solar state using contemporaneous solar-monitor information.

Only observations meeting the required solar-flux and particle-background quality criteria should contribute to quantitative elemental inversion.

9.2 Energy Calibration

The detector energy scale is represented by

$$E(p) = b_0 + b_1 p + b_2 p^2 + \dots, \quad (26)$$

where p is detector channel.

Energy drift should be tracked with time and instrument state.

9.3 Line Fitting

The observed spectrum can be represented as

$$C(E) = B(E) + \sum_k A_k L_k(E) + P(E), \quad (27)$$

where $B(E)$ is the continuum/background, $L_k(E)$ are instrument-broadened fluorescence lines, and $P(E)$ represents particle-induced components.

The fitting process should return both line fluxes and their covariance matrix.

9.4 Fundamental-Parameter Inversion

Elemental abundance is inferred from the measured line intensities using a physical XRF response model,

$$\mathbf{F} = \mathcal{G}(\mathbf{c}, \mathbf{S}, \mathbf{G}, \mathbf{R}_{det}). \quad (28)$$

The inverse solution is

$$P(\mathbf{c} | \mathbf{F}) \propto P(\mathbf{F} | \mathbf{c})P(\mathbf{c}). \quad (29)$$

10 Spatial Harmonization of IIRS and CLASS

IIRS and CLASS operate at substantially different spatial scales. Direct pixel-to-pixel comparison is therefore invalid.

For a CLASS footprint k , aggregate valid IIRS measurements using uncertainty and quality weights:

$$\bar{P}_{\text{IIRS},k} = \frac{\sum_i w_i P_i}{\sum_i w_i}. \quad (30)$$

Weights should incorporate:

$$w_i \propto \frac{q_i}{\sigma_i^2}, \quad (31)$$

where q_i is a quality/homogeneity factor.

The aggregation should also test lithological homogeneity. A CLASS footprint containing multiple lithologies should not be treated as a single pure calibration target.

The covariance generated by spatial aggregation must be retained.

11 Multimodal Fusion

11.1 Bayesian Fusion

The central fusion equation is

$$P(\mathbf{c} \mid \mathbf{y}_{NIR}, \mathbf{y}_{XRF}) \propto P(\mathbf{y}_{NIR} \mid \mathbf{c})P(\mathbf{y}_{XRF} \mid \mathbf{c})P(\mathbf{c}). \quad (32)$$

The NIR likelihood constrains mineralogical and chemical consistency, while the XRF likelihood anchors elemental abundance.

11.2 Physics-Constrained Neural Fusion

A neural architecture may define

$$\mathbf{z}_{NIR} = f_{\theta}(R_{IIRS}), \quad (33)$$

$$\mathbf{z}_{XRF} = g_{\phi}(F_{Mg}, F_{Al}, F_{Si}, F_{Ca}, F_{Fe}), \quad (34)$$

and

$$\mathbf{z} = [\mathbf{z}_{NIR}, \mathbf{z}_{XRF}, \mathbf{G}]. \quad (35)$$

The final elemental prediction is

$$\hat{\mathbf{c}} = h_{\psi}(\mathbf{z}). \quad (36)$$

The loss function should contain observational, physical, stoichiometric, and spatial terms:

$$\mathcal{L} = \mathcal{L}_{spectral} + \lambda_1 \mathcal{L}_{XRF} + \lambda_2 \mathcal{L}_{stoich} + \lambda_3 \mathcal{L}_{physics} + \lambda_4 \mathcal{L}_{spatial}. \quad (37)$$

11.3 Physical Constraints

The network should satisfy

$$f_i \geq 0, \quad \sum_i f_i = 1, \quad (38)$$

and elemental abundances must remain non-negative.

Stoichiometric reconstruction is represented by

$$\mathbf{c}_{NIR} = \mathbf{S}\mathbf{f}, \quad (39)$$

where $\mathbf{S}$ is the mineral stoichiometry matrix.

The predicted composition should also satisfy geochemical closure within the uncertainty associated with unmodeled volatile/light-element components.

12 Uncertainty Quantification

12.1 Measurement Uncertainty

Detector and photon noise should be propagated through the complete calibration chain.

$$\sigma_{\text{shot}} = \frac{\sqrt{N_e + N_{\text{dark}} + RN^2}}{N_e}. \quad (40)$$

12.2 Band-Depth Uncertainty

For

$$BD = 1 - \frac{R_b}{R_c}, \quad (41)$$

the variance is

$$\sigma_{BD}^2 = \frac{\sigma_{R_b}^2}{R_c^2} + \frac{R_b^2 \sigma_{R_c}^2}{R_c^4} - 2 \frac{R_b}{R_c^3} \text{Cov}(R_b, R_c). \quad (42)$$

Correlated calibration errors must therefore not be treated as independent channel noise.

12.3 Monte Carlo Retrieval

For final products, use correlated perturbations:

```
for k = 1 ... N:
    R_k = R + correlated_noise(C_R)
    lambda_k = lambda + noise(sigma_lambda)
    anchors_k = perturb_continuum_anchors()
    Rc_k = fit_continuum(R_k, anchors_k)
    params_k = extract_features(R_k, Rc_k)
    chem_k = nonlinear_inversion(params_k, R_k)

report median and 16/84 percentiles
```

At least $N = 500$ realizations are recommended for routine retrieval and approximately $N = 1000$ for high-value science maps.

12.4 Three-Level Uncertainty Budget

Separate:

1. measurement uncertainty;
2. retrieval uncertainty;
3. geological/model uncertainty.

This prevents model inadequacy from being incorrectly represented as detector noise.

13 Detection Limits and Quality Flags

A conservative detection threshold is

$$BD_{\text{lim}} = 3\sigma_{BD,MC}. \quad (43)$$

If

$$BD < BD_{\text{lim}}, \quad (44)$$

the product should report an upper limit rather than a numerical composition.
Recommended quality flags are:

- GOOD;
- SNR.LIMITED;

- THERMAL_CONTAMINATED;
- SATURATED;
- MIXTURE_AMBIGUOUS;
- DETECTION_LIMIT;
- OUTSIDE_CALIBRATION.

14 Laboratory and External Validation

14.1 Laboratory Spectral Libraries

Candidate libraries include RELAB, USGS spectral libraries, and HOSERLab measurements.

The laboratory spectra should be transformed into flight-like observations by:

1. spectral convolution to IIRS resolution;
2. wavelength-grid resampling;
3. noise injection;
4. grain-size variation;
5. space-weathering variation;
6. thermal and photometric perturbation where appropriate.

The pipeline should then recover the known mineral chemistry within the declared uncertainty.

14.2 Cross-Instrument Validation

The validation hierarchy should include:

1. IIRS internal calibration and processing validation;
2. M³ spectral overlap;
3. laboratory mineral spectra;
4. CLASS-XRF elemental products;
5. lunar sample geochemistry;
6. orbital gamma-ray measurements for independent regional-scale validation;
7. independent mineralogical maps.

14.3 Blind Spatial Validation

Random pixel splitting should be avoided because neighboring lunar pixels are spatially correlated. Instead, use geographically separated training, validation, and test regions:

TRAIN
- geological region A
- geological region B
- geological region C
VALIDATION
- independent region D
TEST
- geographically separated region E

The test region must not participate in calibration or hyperparameter selection.

15 Model Comparison and Ablation Study

The research should benchmark increasingly sophisticated models:

Table 2: Recommended model hierarchy.

Model	Scientific purpose
Band-center regression	Classical spectroscopy baseline
PLSR	Linear statistical baseline
Random Forest / XGBoost	Nonlinear statistical baseline
Linear spectral unmixing	Mineralogical baseline
Hapke/Shkuratov inversion	Physics-based baseline
Shkuratov + ML surrogate	Scalable nonlinear baseline
Physics-constrained neural inversion	Proposed advanced model
IIRS + CLASS multimodal fusion	Final research model

Ablation studies should remove one component at a time:

- no thermal correction;
- no weathering parameter;
- no grain-size parameter;
- linear instead of nonlinear mixing;
- no CLASS information;
- no spatial harmonization;
- no physics constraints.

This demonstrates which methodological components actually improve retrieval accuracy.

16 Closure and Sanity Tests

16.1 Synthetic Injection-Recovery

Known absorptions should be injected into representative spectra across SNR values of approximately 10–200.

The retrieval should satisfy:

$$|\Delta\lambda_c| < \sigma_\lambda \quad (45)$$

and, for statistically significant bands,

$$\frac{|\Delta BD|}{BD} < 10\%. \quad (46)$$

16.2 Continuum Sensitivity

Reprocess scenes with:

- convex hull;
- second-order polynomial;
- perturbed anchor points.

If the composition changes by more than the reported uncertainty, the method uncertainty must be inflated.

16.3 Spatial Closure

Adjacent orbital observations should agree within combined uncertainty. Track-boundary steps indicate residual calibration or photometric errors.

16.4 Geochemical Closure

Reconstructed oxide abundances should satisfy an approximate closure condition,

$$\sum_i X_i = 100 \pm \delta \text{ wt.}\%, \quad (47)$$

where δ explicitly accounts for excluded volatile/light-element components and model uncertainty.

17 Final Data Products

The proposed processing levels are:

- L1:** calibrated radiance / I/F with detector, wavelength, and calibration uncertainties;
- L2:** thermal- and photometrically corrected reflectance plus spectral parameters;
- L3:** mineralogical retrieval including modal fractions, Mg#, Fo#, En#, Wo#, FeO proxy, grain size, and maturity posterior;
- L4:** CLASS-calibrated elemental ratios Mg/Si, Al/Si, Ca/Si, and Fe/Si;
- L5:** multimodal geochemical maps with posterior uncertainty, detection limits, quality flags, and provenance.

Every L4/L5 elemental-ratio pixel should contain:

- posterior median;
- 1σ or 68% credible interval;
- 95% credible interval;
- detection limit;
- quality flag;
- calibration version;
- number of Monte Carlo samples;
- continuum method;
- spectral library version;
- CLASS observation/solar-state metadata;
- spatial aggregation footprint;
- validation status.

18 Research Hypothesis and Expected Contribution

The central research hypothesis is:

A physics-constrained multimodal inversion combining high-spatial-resolution lunar NIR spectroscopy with lower-spatial-resolution but compositionally direct XRF measurements can retrieve elemental ratios with lower systematic bias and better quantified uncertainty than either NIR-only empirical regression or CLASS-only interpolation.

The principal scientific contribution is therefore not simply an improved regression equation. It is a framework in which:

$$\boxed{\text{IIRS mineralogical information} + \text{CLASS elemental information} + \text{physical forward modeling} + \text{uncertainty-a}} \quad (48)$$

produces spatially resolved and physically interpretable lunar geochemical products.

19 Implementation Blueprint

INPUT - IIRS Level-1 radiance - CLASS XRF spectra/events - SPICE geometry - DEM/topography - solar-state data - laboratory spectral libraries IIRS PIPELINE - detector calibration

wavelength calibration
bad-pixel masking
thermal correction
photometric correction
standardized reflectance

CLASS PIPELINE

energy calibration
solar-state screening
particle-background subtraction
detector-response correction
XRF line fitting
elemental inversion

SPECTRAL PIPELINE

continuum removal
band parameters
full-spectrum embedding
nonlinear spectral unmixing
grain-size retrieval
space-weathering retrieval
mineral chemistry

SPATIAL PIPELINE

co-registration
IIRS -> CLASS footprint aggregation
homogeneity testing
covariance propagation

FUSION

Bayesian inversion OR
physics-constrained neural model

OUTPUT

mineral chemistry
Mg/Si
Al/Si
Ca/Si
Fe/Si
uncertainty posterior
detection limits
quality flags
provenance

VALIDATION

laboratory recovery
synthetic injection-recovery
M3/IIRS comparison
CLASS comparison
lunar sample comparison
independent regional blind test
geochemical closure

20 Selected Research Literature

References

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